

Game Strategy and Goals Menu

What Should Our Robot Try to Do in a Match?

Graduate Expectation Tags

- **[Red: Critical Thinking & Problem Solving]** We compare scoring, parking, reliability, and penalties before choosing a strategy.
- **[Blue: Effective Communication]** We explain our match plan clearly so drivers, human operators, builders, programmers, and presenters understand it.
- **[Purple: Curiosity & Creativity]** We imagine different ways to score, park, and work with an alliance partner.
- **[Green: Personal Development]** We choose a realistic goal based on time, materials, skill, and evidence.
- **[Orange: Cultural Understanding & Civic Engagement]** We think about alliance teamwork, field space, and responsible competition.

Decision Being Made

Your team is choosing its first match strategy. A strategy is not just what the robot can do. It is what the team will practice, test, explain, and improve.

In this course, your robot will not use an intake. A human operator will load balls into the robot. Your team will design a hopper that can carry up to three balls and a launching system that can score those balls.

Your team also has limited time and materials. You do not have sensors for reading AprilTags or measuring exact distance, so aiming and driving will depend on driver practice, field marks, repeatable starting positions, and simple autonomous movements.

Background Sources

Use ideas from:

- **What Is FIRST Tech Challenge?**
- **Robotics Safety Norms**
- **Borrowed Equipment and Community Trust**
- **Chassis Shape Menu**
- **Wheel Selection Menu**
- **Motor Placement Menu**
- **Electronics and Battery Placement Menu**
- **DECODE Scoring and Strategy Background**
- drive practice, field observations, and early chassis tests
- the current game rules or course scoring summary

Why This Decision Matters

Teams often want to build every possible scoring feature. In a short course, that can make the robot late, fragile, or hard to drive. A strong strategy helps the team decide what matters most.

A good strategy answers questions like:

- Where should the robot launch from?
- How many scoring trips should the team attempt?
- How will the human operator load balls quickly and safely?
- How will the robot avoid penalties?
- How will the robot leave room for an alliance partner?
- Should the team prioritize reliable parking, scoring, or an ambitious challenge?
- What evidence will show whether the strategy is working?

There is no perfect strategy. Your job is to choose a strategy that your team can build, test, drive, and explain with evidence.

Important Course Constraints

These are not menu options. They are constraints your team must design around.

Constraint	What It Means for Strategy
No intake	The robot will not collect balls from the field. A human operator will load balls.
Three-ball maximum hopper goal	The hopper should hold up to three balls so the robot can attempt a scoring run after loading.
No AprilTag or distance sensors	Aiming and autonomous movement must rely on driver judgment, field marks, timing, encoder estimates, or repeatable starting positions.
Limited build time	A reliable simple strategy may score more than a complex unfinished strategy.
Shared field space	The robot should avoid blocking an alliance partner unless that is part of an agreed strategy.
Penalties matter	Avoiding penalties can be as important as scoring more points.

Menu Options

Option A: Reliable Three-Ball Scoring Run

Graduate Expectation Connection: [Red: Critical Thinking & Problem Solving]

What it means: The robot is designed to load up to three balls, drive to a chosen launch site, aim by driver judgment or field marks, and score the three balls as reliably as possible.

What it helps: This keeps the strategy focused. The team can practice one repeatable scoring path and collect evidence about accuracy, cycle time, and driver control.

Best for: Teams that want a clear first scoring goal and a robot that is easier to build, test, and present.

Ease of use: Medium

Likelihood of success: Medium to high

Time to implement: Moderate

Possible trade-offs: The team may score fewer total balls than a faster cycle strategy. If the launch site is blocked or defended, the robot needs a backup plan.

Evidence we might watch: How many of three balls score from the chosen site? How long does one loading-driving-shooting cycle take? Can different drivers repeat the path?

Option B: Two or Three Short Scoring Cycles

Graduate Expectation Connection: [Purple: Curiosity & Creativity] and [Red: Critical Thinking & Problem Solving]

What it means: The team tries to make multiple trips: load, drive, score, return, reload, and score again. The goal is to score more than the first three balls.

What it helps: This can increase scoring if the robot drives reliably, the human operator loads quickly, and the launch site is easy to reach.

Best for: Teams with a reliable chassis, quick loading plan, and drivers who can repeat a route without losing control.

Ease of use: Medium to hard

Likelihood of success: Medium

Time to implement: Moderate to slow

Possible trade-offs: More cycles create more chances for missed shots, traffic, defense, loading delays, or penalties. The team may spend too much time chasing speed before the first scoring cycle is reliable.

Evidence we might watch: Can the robot complete two full cycles before time runs out? Does the second cycle score as reliably as the first? Where does the team lose the most time?

Option C: Specific Launch Site Strategy

Graduate Expectation Connection: [Blue: Effective Communication] and [Red: Critical Thinking & Problem Solving]

What it means: The team chooses one legal launch site and designs driving practice, aiming, hopper position, and shooter angle around that location.

What it helps: A known launch site can make aiming more repeatable. It also helps the team communicate with an alliance partner about field space.

Best for: Teams that need consistency because they are aiming without distance sensors or AprilTag reading.

Ease of use: Medium

Likelihood of success: High if the site is legal, reachable, and repeatable

Time to implement: Fast to moderate

Possible trade-offs: The site may be blocked by another robot, a field object, or defense. A strategy built around one spot needs a backup site or recovery plan.

Evidence we might watch: Can the robot reach the site without hitting field elements? Does the robot score more accurately from that site than from other sites? Can the alliance partner still move and park?

Option D: Alliance-Friendly Parking Strategy

Graduate Expectation Connection: [Orange: Cultural Understanding & Civic Engagement]

What it means: The robot is designed and driven so both alliance robots can fully park or fit in the required end area when the match ends.

What it helps: This may earn extra endgame, ranking, or advancement value if the current rules reward both alliance robots parking. It also shows strong alliance awareness.

Best for: Teams that want a reliable endgame contribution and are willing to design around shared space.

Ease of use: Medium

Likelihood of success: Medium to high if practiced

Time to implement: Moderate

Possible trade-offs: A narrow robot or unusual shape may leave less room for hopper, launcher, battery, electronics, or stability. A tall robot that allows another robot underneath may be risky, harder to build, and harder to keep stable.

Evidence we might watch: Can two robots fully fit in the parking area? Can the robot park quickly after scoring? Does the robot remain stable if it is tall or narrow?

Option E: Simple Autonomous Mobility

Graduate Expectation Connection: [Green: Personal Development] and [Red: Critical Thinking & Problem Solving]

What it means: The robot has at least one autonomous program that drives away from the starting area or off the back in a predictable way.

What it helps: This gives the team a realistic autonomous goal. It can be tested early and improved without requiring sensors or complex aiming.

Best for: Teams that need a dependable autonomous routine before attempting autonomous scoring.

Ease of use: Easy to medium

Likelihood of success: High with testing

Time to implement: Fast to moderate

Possible trade-offs: This may score fewer autonomous points than a scoring routine, but it is more likely to work consistently.

Evidence we might watch: Does the robot end in the correct area three times in a row? Does battery level or starting position change the result? Can the team explain the program?

Option F: Autonomous Scoring Challenge

Graduate Expectation Connection: [Purple: Curiosity & Creativity] and [Red: Critical Thinking & Problem Solving]

What it means: The team attempts to score during autonomous using a repeatable starting position, timed movement, motor power, encoder estimates, or a fixed launch setup.

What it helps: This is an ambitious challenge that can create strong portfolio evidence if the team tests carefully and explains what worked or failed.

Best for: Teams that already have reliable basic autonomous movement and a launcher that scores consistently from a known setup.

Ease of use: Hard

Likelihood of success: Risky

Time to implement: Slow

Possible trade-offs: Autonomous scoring may consume time needed for driver practice, hopper reliability, or basic scoring. Without sensors, small changes in starting position may cause misses.

Evidence we might watch: How often does autonomous scoring work from the same starting setup? What variable causes misses: starting angle, launch speed, battery level, ball position, or robot drift?

Option G: Penalty-Avoidance Strategy

Graduate Expectation Connection: [Green: Personal Development] and [Orange: Cultural Understanding & Civic Engagement]

What it means: The team makes avoiding penalties a major part of the match plan. Drivers, human operators, and coaches practice legal loading, legal launching, safe contact, and field awareness.

What it helps: Avoiding penalties protects the team's score and shows discipline. A clean, reliable match may beat a risky match with more attempted scoring.

Best for: All teams, especially teams with new drivers or a robot that will operate near other robots.

Ease of use: Medium

Likelihood of success: High with practice

Time to implement: Fast to moderate

Possible trade-offs: The team may drive more cautiously and attempt fewer shots while learning the rules.

Evidence we might watch: How many practice runs are completed without rule warnings? Can the driver and human operator explain the riskiest penalty situations?

Strategy Comparison Table

Strategy	Best Feature	Main Risk	Good For
Reliable three-ball scoring run	Clear and testable	May limit total scoring	First scoring plan
Two or three short scoring cycles	Higher scoring potential	More chances for delays and mistakes	Fast, reliable drive teams
Specific launch site	Repeatable aiming	Site may be blocked	Sensor-free scoring
Alliance-friendly parking	Helps alliance endgame	May constrain robot shape	Teams designing for shared space
Simple autonomous mobility	Reliable first autonomous	Lower ambition	Early autonomous success
Autonomous scoring challenge	Strong stretch evidence	Risky without sensors	Advanced testing teams
Penalty avoidance	Protects score	May feel cautious	All teams

Strategy Decision Table

Record your decision in your engineering portfolio.

Decision Area	Our Choice	Why We Chose It	What We Gave Up	Evidence We Will Watch
Main scoring strategy				
Launch site				
Number of scoring cycles				
Parking plan				
Autonomous goal				
Penalty-avoidance focus				

Launch Site Planning

Choose one first launch site and one backup launch site.

Launch Site	Why It Might Work	What Could Block It	Evidence to Test
First choice			
Backup choice			

Before finalizing a launch site, check:

- Is the site legal under the current rules?
- Can the robot reach the site safely?
- Can the human operator load the robot before it drives there?
- Can the driver aim from that site without sensors?
- Can an alliance partner still move, score, or park?
- What should the robot do if the site is blocked?

Must / Should / Could

Level	Expectation
Must Do	Choose a main scoring strategy, one legal launch site, and one autonomous mobility goal.
Should Do	Explain the main trade-off in the strategy and identify evidence the team will collect during testing.
Could Do	Compare two strategies using cycle time, scoring accuracy, parking space, penalty risk, or alliance impact.

SLD Connection

Main SLD Prompt

Should our team prioritize reliable three-ball scoring, multiple scoring cycles, alliance-friendly parking, or autonomous challenge points?

Supporting Questions

- What can our robot realistically do with no intake?
- How many balls can we score reliably from one loading cycle?
- Where is our best legal launch site?
- What launch site still leaves room for an alliance partner?
- Should our robot shape help two alliance robots park?
- What is the safest first autonomous goal?
- When would autonomous scoring be worth the extra risk?
- What penalties are most important for us to avoid?
- What evidence should we collect before changing our strategy?

Portfolio Deliverable

Add the following to your engineering portfolio:

- selected match strategy,
- chosen launch site and backup launch site,
- hopper goal and expected number of balls per scoring run,
- expected number of scoring cycles,
- autonomous goal,
- parking plan,
- penalty-avoidance focus,
- one major trade-off,
- evidence the team will collect during testing.

Reflection

Our main strategy is:

We chose this strategy because:

The main trade-off we accepted is:

The evidence we will collect is:
